

# Supplementary Appendix for Temporal Hand Labanotation

Anonymous submission

## Purpose

This appendix records the concrete artifacts, protocols, and settings used by the anonymous submission. It is intended as a reproducibility companion to the main paper and a claim-to-evidence map for the reported results.

## Directory Layout

All paths below are relative to the project root unless stated otherwise.

Submission package root:

`paper/submission_ready/`

Main paper file:

`paper/submission_ready/`

`AnonymousSubmission2027.tex`

Checklist file:

`paper/submission_ready/`

`ReproducibilityChecklist.tex`

Generated experiment artifacts:

`experiments/generated/`

Executable scripts:

`tools/*.py`

Base dataset root:

`/opt/tiger/InterHand`

## Claim-to-Evidence Map

Claim 1: THL improves symbolic structure over a flat HL-style sequence representation.

- Main-paper evidence: Table 1A and the ablation table
- Source artifacts: `experiments/generated/topline_evidence_bundle.json` and `experiments/generated/temporal_representation_ablation_report.json`

Claim 2: THL yields substantially stronger local editability than matched proxy token spaces.

- Main-paper evidence: Table 1A

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

- Source artifacts: `experiments/generated/local_edit_audit.json`, `experiments/generated/learned_token_proxy_report.json`

Claim 3: THL improves interaction-conditioned preservation under the same held-out protocol.

- Main-paper evidence: Table 1B and the statistical discussion in Results
- Source artifacts: `experiments/generated/topline_evidence_bundle.json`, `experiments/generated/interaction_realized_significance_report.json`, `experiments/generated/interaction_realized_feasible_left_repair_bundle.json`

Claim 4: Short transition chunks are the strongest minimal temporal upgrade.

- Main-paper evidence: Table 1C and the ablation table
- Source artifacts: `experiments/generated/interaction_realized_closing_chunk_transfer_val_only.json`, `experiments/generated/interaction_realized_opening_chunk_transfer_val_only.json`, `experiments/generated/temporal_representation_ablation_report.json`

The ablation table should be read within the learned sequence-model family rather than as a direct replacement for every value in the main result table. The main result table mixes structure, editability, and interaction-preservation audits, whereas the ablation isolates which temporal channels matter inside the matched sequence-model comparison.

## Temporal HL Export And Preprocessing

The submission does not introduce a new raw image dataset. Instead, it reuses the HL-derived symbolic export already built from InterHand2.6M and related annotations in the current workspace.

Temporal export entry point:

`tools/build_temporal_hl.py`

Train split used in symbolic pretraining:  
experiments/generated/temporal\_hl\_train.json  
Validation split used for support construction and tuning:  
experiments/generated/temporal\_hl\_val.json  
Test split used for final evaluation:  
experiments/generated/temporal\_hl\_test.json

The main preprocessing stages are:

- conversion from joint annotations to symbolic hand states
- temporal windowing for learned sequence models
- support-bank construction for symbolic intervention and chunk-transfer audits
- slice filtering for hard interaction and feasible-subset evaluations

## Main Paper Numbers And Their Source Artifacts

Topline evidence bundle used for Table 1:  
experiments/generated/topline\_evidence\_bundle.json

Ablation report used for Table 2:  
experiments/generated/temporal\_representation\_ablation\_report.json

Locality audit:  
experiments/generated/local\_edit\_audit.json

Interaction significance report:  
experiments/generated/interaction\_realized\_significance\_report.json

Feasible interaction repair bundle:  
experiments/generated/interaction\_realized\_feasible\_left\_repair\_bundle.json

Broader closing chunk-transfer report:  
experiments/generated/interaction\_realized\_closing\_chunk\_transfer\_val\_only.json

Broader opening chunk-transfer report:  
experiments/generated/interaction\_realized\_opening\_chunk\_transfer\_val\_only.json

## Main-Paper Values

The paper reports the following principal values from the artifacts above.

- structure: flat symbolic sequence 0.6923 vs grouped symbolic sequence 0.8974
- locality: symbolic locality 0.3553 vs semantic proxy 0.0945 and continuous proxy 0.0848
- hard interaction, strict full slice: closing/opening 0.0250/0.0533

- hard interaction, corrected support: closing/opening 0.1807/0.1812
- feasible interaction subset: closing/opening 0.6392/0.6497
- preserve-side repair: closing/opening 0.6582/0.6497
- chunk transfer: closing/opening 0.7785/0.7197 against fixed-edge 0.7025/0.6306

## Protocol Fairness Notes

The central comparisons in the main paper are designed to isolate the effect of the representation rather than the effect of a stronger downstream solver. The fairness conditions used throughout are:

- the symbol inventory is unchanged when THL is compared against flat HL-style symbolic baselines
- the downstream operator family is held fixed within each audit (local intervention, interaction-conditioned preservation, or chunk transfer)
- support banks are built from validation data rather than test data
- final reporting is performed on held-out unseen-subject test splits
- learned-token controls use matched split families and stronger training budgets rather than weaker geometric heuristics

This means the reported gains should be interpreted as gains in notation structure and editability under matched operator strength, not as gains from a privileged retrieval or repair mechanism.

## Statistical Evidence

The paper reports significance only for the interaction-conditioned frontier, where frame-level variance is most consequential. The corresponding artifact is experiments/generated/interaction\_realized\_significance\_report.json.

The reported results are:

- on the full hard slice, corrected support improves closing by +0.156 with 95% CI [0.127, 0.186] and opening by +0.128 with 95% CI [0.101, 0.156]; paired permutation tests are below  $10^{-4}$  in both cases
- on the feasible interaction subset, preserve-side repair improves closing by +0.184 with 95% CI [0.108, 0.259] and opening by +0.166 with 95% CI [0.108, 0.223]; the corresponding  $p$ -values are  $5 \times 10^{-5}$  and below  $10^{-4}$

The sequence-accuracy comparisons additionally use three random seeds  $\{0, 1, 2\}$  for learned sequence models, while symbolic intervention and chunk transfer are deterministic once the split and support bank are fixed.

## Claim Boundary

The submission makes a deliberately narrow claim. It does not claim that THL is a universal solution for every downstream hand-motion task, nor that the paper

exhausts every possible evaluation of symbolic hand notation. The supported claim is instead:

- HL can be upgraded from framewise notation into a temporal symbolic motion representation
- this upgrade improves structure, local editability, and interaction-conditioned preservation under matched held-out protocols
- the strongest minimal upgrade is grouped state plus short transition motifs, rather than heavier temporal channels

This is the scope the main paper is written to defend, and the artifacts listed above are selected accordingly.

### Protocol Sketch

For clarity, the overall evaluation protocol used in the paper can be summarized as follows.

1. Build temporal symbolic train/validation/test splits from the HL-derived export.
2. Train the learned sequence models on the train split and select hyperparameters on validation.
3. Build support banks only from validation data.
4. Keep the downstream operator family fixed for the comparison being audited.
5. Evaluate on the held-out test split, including unseen-subject slices and the interaction-conditioned subsets used in the main paper.

This sketch is intentionally high level: the exact entry points and artifact files are listed below, while the main paper focuses on the representation claim rather than on engineering workflow detail.

### Common Reviewer Checks

The present submission is designed to rule out three common alternative interpretations.

- “This is just a stronger recognizer.” The paper does not evaluate THL only through framewise prediction; it evaluates structure, locality, and interaction-conditioned preservation under fixed downstream operator classes.
- “The gain comes from a larger symbol set.” The grouped state component preserves the original HL symbol inventory; the measured gain comes from reorganizing the object and adding short transition motifs.
- “The gain comes from a privileged solver.” Support banks, operator families, and held-out split protocols are matched across the comparisons used for the main claims.

### Final Hyperparameters Reported In The Main Paper

Learned temporal symbolic sequence models.

- seeds: {0, 1, 2}
- window size: 32

- stride: 16
- hidden dimension: 128
- pretraining epochs: 20
- finetuning epochs: 20

Matched learned-token controls.

- compared token codebooks: 32 and 64
- final reported codebook size: 64
- max codebook frames: 100,000
- pretraining window span: 186
- pretraining step: 96
- pretraining epochs: 120
- finetuning epochs: 120

Chunk transfer.

- tested chunk lengths: 2, 3, 4
- broader opening result is stable across all three lengths
- broader closing result is reported at the strongest tested setting

### Representative Commands

Matched old-HL vs temporal-HL boundary check. Command:

```
python tools/build_oldhl_temporal_matched_report.py
```

Grouped/repair structural analysis. Command 1:

```
python tools/build_current_code_symbolic_frontier.py
```

Command 2:

```
python tools/build_representation_risk_bundle.py
```

Broader feasible closing chunk transfer. Command:

```
python tools/build_interaction_realized_closing_chunk_transfer.py
```

Broader feasible opening chunk transfer. Command:

```
python tools/build_interaction_realized_opening_chunk_transfer_val_only.py
```

Sequence-model training family. Command 1:

```
python tools/train_symbolic_pretrain.py
```

Command 2:

```
python tools/train_symbolic_pretrain_grouped.py
```

Command 3:

```
python tools/train_joint_token_baseline.py
```

### Script Inventory Used Directly By The Submission

The submission relies primarily on the following executable entry points.

Entry point 1:

```
tools/build_temporal_hl.py
```

Entry point 2:

```
tools/build_current_code_symbolic_frontier.py
```

Entry point 3:  
tools/build\_representation\_risk\_bundle.py  
Entry point 4:  
tools/build\_toplevel\_evidence\_bundle.py  
Entry point 5:  
tools/build\_oldhl\_temporal\_matched\_report.py  
Entry point 6:  
tools/build\_interaction\_realized\_closing\_chunk\_transfer.py  
Entry point 7:  
tools/build\_interaction\_realized\_opening\_chunk\_transfer\_val\_only.py  
Entry point 8:  
tools/train\_symbolic\_pretrain.py  
Entry point 9:  
tools/train\_symbolic\_pretrain\_grouped.py  
Entry point 10:  
tools/train\_joint\_token\_baseline.py  
Entry point 11:  
tools/build\_paper\_figures.py

to understand or reproduce the reported anonymous results.

### Selection Criteria

The final settings used in the main paper were not chosen post hoc from all available experiments. We used the following rule:

- for matched learned controls, keep the setting with the strongest validation-side sequence accuracy under the same split protocol
- for chunk transfer, keep the setting with the strongest corrected feasible-slice score, but only if the effect remains stable across the tested chunk lengths or support-bank variants
- keep only evidence that directly supports the paper’s central representation claim under the held-out protocol

### Environment

The current environment used to generate the final paper bundle is:

- Python 3.11.2
- PyTorch 2.9.1
- CUDA-enabled execution path in the current training scripts
- NVIDIA A100 80GB PCIe GPUs in the active machine environment

### Anonymity And Scope

The submission package contains only anonymous paper assets, bibliography files, figures, checklist material, and this appendix. Internal rendering snapshots, exploratory debug images, and unrelated workspace artifacts are not part of the final submission bundle. The present appendix documents only the artifacts required